

Pose-Aware Multi-Agent Deep Reinforcement Learning for Football Player Action Valuation

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Abstract. Player posture provides cues about imminent actions and movement preparation that are not fully captured by position and velocity alone. We present a pose-aware multi-agent deep reinforcement learning (MADRL) framework for the valuation of football players’ actions. A SARSA-based action-value model is augmented with 3D pose features, and three state representations are compared: no pose, raw joint coordinates, and a latent representation learned with a Spatial-Temporal Graph Convolutional Network (ST-GCN). Experiments on UEFA Champions League tracking, event, and pose data show that raw pose features improved balanced accuracy from 0.0836 to 0.1637, while macro-F1 decreased slightly despite a lower total loss. The current latent ST-GCN representation does not improve over the no-pose baseline, suggesting that pose information can be useful but its representation remains critical under limited data.

Keywords: Multi-agent · Reinforcement learning · 3D pose · Pose embedding · Sports · Football

1 Introduction

Football analytics has increasingly relied on event and tracking data to model player behavior and team tactics. Event data describe discrete on-ball actions such as passes, shots, and defensive events, while tracking data provide continuous information about the spatial configuration of players and the ball. Together, these data sources have enabled models that estimate the value of actions [3, 13, 14], possessions [12, 18], and tactical patterns [1, 10] beyond traditional box-score statistics.

A particularly important direction is action valuation, which aims to estimate the contribution of each player’s action to future scoring opportunities or match outcomes. Previous studies have proposed value-based frameworks such as expected possession value [4, 9] and reinforcement-learning-based action valuation [6, 7, 11]. In particular, multi-agent deep reinforcement learning has been used to model football as a sequential decision-making problem involving multiple interacting players, allowing the valuation of both on-ball and off-ball actions [7]. These approaches are well-suited to football because the value of an action

depends not only on the ball carrier but also on the positioning and movement of teammates and opponents.

However, most existing action-valuation models rely primarily on player and ball positions and velocities. Although these variables capture where players are and how they move, they do not fully describe their physical preparedness to act. Body posture may contain additional information about a player’s orientation, balance, readiness, and immediate action affordances. For example, two players may have similar locations and speeds, but one may be facing forward and prepared to receive or pass, while the other may be off balance or oriented away from the ball. Such differences may affect both the probability of future actions and their expected value.

Recent advances in player pose estimation enable the incorporation of three-dimensional body-joint information into football analytics [2, 8, 15, 16, 19]. Nevertheless, their use in value-based decision models for football remains limited. In particular, it is unclear how pose information should be represented and integrated into multi-agent action-valuation models. Raw joint coordinates may preserve detailed posture information but increase the dimensionality of the state representation. In contrast, learned pose embeddings, such as those obtained from spatiotemporal graph convolutional networks [17], may provide compact representations of movement patterns but may discard information relevant to downstream action valuation, especially with limited training data.

This study investigates whether 3D pose information improves the valuation of football players’ actions in a multi-agent deep reinforcement learning framework. We augment a SARSA-based action-value model [7] with pose features and compare three state representations: a baseline representation without pose, a raw-pose representation using 3D joint coordinates, and a latent pose representation learned with a Spatial-Temporal Graph Convolutional Network (ST-GCN) [17]. Using UEFA Champions League tracking, event, and pose data, we evaluate these representations with respect to action prediction and value-estimation losses.

The main contributions of this paper are as follows. First, we introduce a pose-aware extension of MADRL-based football action valuation. Second, we compare raw joint coordinates and learned pose embeddings as state representations within the same reinforcement learning framework. Third, we provide an empirical comparison suggesting that the effectiveness of pose information depends strongly on its representation, with raw joint coordinates outperforming the learned ST-GCN embedding in the current experimental setting.

2 Methods

2.1 Dataset and preprocessing

We used UEFA Champions League 2022/23 group-stage match data, including event data, player and ball tracking data, and three-dimensional player pose coordinates. Tracking and pose data were recorded at 25 Hz, and the event data

were synchronized with them using ELASTIC [5]. We followed the preprocessing pipeline provided by OpenSTARLab [20].

The coordinate system was aligned so that the attacking direction was consistently left-to-right. For both 2D tracking and 3D pose data, the origin and direction were standardized during preprocessing. The x - and y -coordinates were flipped when necessary so that the attacking team always moved from left to right. We did not apply statistical normalization such as z-score or min-max scaling.

For reinforcement learning, tracking variables were downsampled from 25 Hz to 5 Hz to reduce temporal redundancy and computational cost. Each 5 Hz time step corresponded to a bin of five consecutive 25 Hz frames. Tracking variables were aggregated for the RL state, whereas pose joints were retained as an ordered list of frames within each bin. Pose missingness was handled in two steps. Sequences containing explicitly missing pose frames were removed before training, whereas missing joint entries that remained after this filtering were filled with zeros during feature construction.

We used 12 matches for training, one match for validation, and one match for testing.

2.2 MADRL-based action valuation

We followed the MADRL-based football action-valuation model [7]. This model formulates football action valuation as a sequential decision-making problem and estimate the action values of both on-ball and off-ball players.

At each time step t , the model receives a state s_t , observes an action a_t performed by the target player, and receives a reward r_t . The goal is to estimate the action-value function $Q(s_t, a_t)$, which represents the expected future return after the target player performs action a_t in state s_t . Following SARSA, the temporal-difference target is defined as

$$y_t = r_t + \gamma Q(s_{t+1}, a_{t+1}), \quad (1)$$

where s_{t+1} and a_{t+1} denote the next state and the next observed action, respectively, and γ is the discount factor. We set $\gamma = 1.0$, following the baseline setting.

In the baseline model, the state s_t consists of the two-dimensional positions and velocities of the ball and all players, resulting in a 92-dimensional state vector. The action a_t is represented as one of 16 discrete classes: pass, through pass, shot, cross, dribble, defensive actions such as tackle and interception, movements in eight directions, and stay. The reward r_t follows the previous MADRL framework. Rewards are set to zero for all time steps except the final time step of an attacking sequence. At the final time step, the reward is set to 1.0 if the final action is a shot that results in a goal. If the team concedes a goal in the opponent's attacking sequence immediately after the current sequence, the reward is set to -1.0. In all other cases, the reward is defined using expected possession value [4].

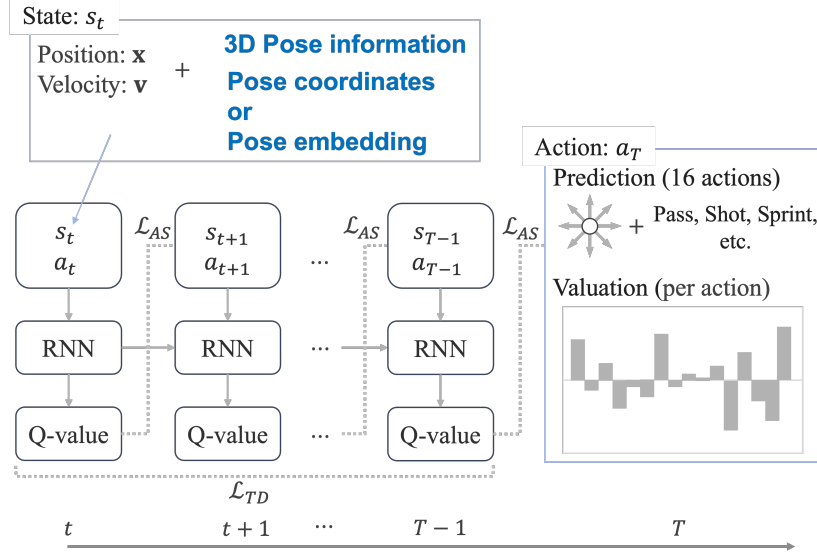


Fig. 1. Overview of the proposed framework. The baseline tracking state is augmented with the target player’s pose information, represented either as raw 3D joint coordinates or as a pose embedding.

The model is trained using a loss function consisting of a temporal-difference loss (TD loss) $\mathcal{L}_{TD}$, an action-prediction loss $\mathcal{L}_{AS}$, and an L1 regularization loss $\mathcal{L}_{L_1}$. The TD loss is defined as

$$\mathcal{L}_{TD} = (y_t - Q(s_t, a_t))^2. \quad (2)$$

The action-prediction loss $\mathcal{L}_{AS}$ is computed as the cross-entropy between the predicted action distribution and the observed action label of the target player. Specifically, the $\mathcal{L}_{AS}$ is defined as

$$\mathcal{L}_{AS} = - \sum_{t \in T} \mathbf{a}_t \cdot \log(\text{softmax}(\mathbf{Q}_{s_t})) \quad (3)$$

where $\mathbf{a}_t \in \{0, 1\}^{|A|}$ is a one-hot action vector, $|A|$ is the number of action classes, and $\mathbf{Q}_{s_t}$ is the vector of action values for all actions at state s_t . The total loss is defined as

$$\mathcal{L} = \mathcal{L}_{TD} + \lambda_1 \mathcal{L}_{AS} + \lambda_2 \mathcal{L}_{L_1}, \quad (4)$$

where λ_1 and λ_2 are weighting coefficients. We set $\lambda_1 = 0.005$ and $\lambda_2 = 0.0001$, following the default setting of the RLearn module in OpenSTARLab [20].

2.3 Pose-aware state representations

Figure 1 provides an overview of the proposed framework. In all model variants, the underlying MADRL action-valuation framework is kept unchanged, and only

the state representation is modified. This design allows us to isolate the effect of pose information on action prediction and value estimation.

The baseline model uses a state representation composed of the two-dimensional positions and velocities of the ball and all players, resulting in a 92-dimensional state vector. While this representation captures the spatial configuration and motion of the game, it does not explicitly encode the target player’s body orientation, limb configuration, or movement readiness.

To incorporate posture information, we considered two pose-aware state representations. In both variants, pose information was used only for the target player whose action was being evaluated. This design focuses on whether the target player’s own posture provides complementary information beyond the spatial tracking features already included in the baseline state.

Raw pose. In the raw-pose model, the target player’s 3D joint coordinates were aggregated within each 5 Hz bin and appended to the baseline state. Specifically, for each target player, we averaged the available 25 Hz pose frames within the corresponding bin for each of the 29 joints after subtracting the midHip joint coordinates in each frame. Each joint was therefore represented by three player-centered values (x, y, z) , resulting in an 87-dimensional pose vector. This vector was concatenated with the 92-dimensional baseline state, yielding a 179-dimensional state vector. This representation preserved joint-level posture information directly without learning a separate pose encoder, while removing absolute translation that was already represented in the tracking state. However, it also increased the dimensionality of the input and may include redundant or noisy information.

Pose ST-GCN. In the pose ST-GCN model [17], the target player’s pose sequence within each 5 Hz bin was encoded into a compact latent representation. The input to the encoder consisted of the first five consecutive 25 Hz pose frames retained in the bin, ordered by frame ID. Before being input to the ST-GCN-based skeleton encoder, pose coordinates were converted to a player-centered representation by subtracting the midHip joint coordinates. The encoder produced a 32-dimensional latent vector, which was concatenated with the 92-dimensional baseline state, resulting in a 124-dimensional state vector. Thus, the pose sequence represented consecutive frames within the current 5 Hz bin, rather than a symmetric temporal window centered at t .

Compared with raw pose, the pose ST-GCN representation was more compact and was designed to capture short-term spatiotemporal structure in the player’s movement. On the other hand, the compression process may discard posture information that was useful for downstream action valuation. By comparing these three state representations within the same MADRL framework [7], we evaluated not only whether pose information was useful, but also how its representation affected action valuation performance. This comparison should be interpreted with some caution, because both pose-aware variants used player-centered coordinates, but they differed in how the pose information was encoded: raw pose directly exposed an averaged 87-dimensional joint vector, whereas ST-GCN compressed a five-frame joint sequence into a 32-dimensional latent rep-

Table 1. Test results for three pose settings.

Model	Bal. Acc.	Macro F1	Action Loss	L1 Loss	TD Loss	Total Loss
Baseline [7]	0.0836	0.0703	2.2337	145.6158	0.000763	0.02649
+ Raw pose	0.1637	0.0658	2.2372	111.3934	0.000722	0.02305
+ Pose ST-GCN	0.0849	0.0686	2.2399	275.9521	0.000635	0.03943

resentation. Thus, the results may reflect the trade-off between direct access to posture cues and learned temporal compression, rather than a difference in coordinate origin.

2.4 Evaluation metrics

We evaluated the models on the held-out test match using both action-prediction and value-estimation metrics. For action prediction, we reported balanced accuracy (macro recall), macro-F1 score, and cross-entropy action loss. Balanced accuracy was computed using the ‘Accuracy’ metric in ‘torchmetrics’ v1.4.0 with ‘task=multiclass’ and ‘average=macro’ in our implementation. Under this setting, the reported values are numerically equivalent to macro recall, rather than conventional sample-level overall accuracy. To improve readability, we refer to this metric as balanced accuracy in the text and abbreviate it as bal. acc. in Table 1. Macro-F1 was used because the action labels were imbalanced and because it gives equal weight to each action class. For value estimation, we reported the TD loss. The L1 regularization term and total loss were also reported to examine the overall optimization behavior of each state representation.

3 Results

3.1 Quantitative Comparison of State Representations

Table 1 shows the quantitative results of the three state representations on the held-out test match. The raw-pose model achieved the highest balanced accuracy, improving from 0.0836 in the baseline to 0.1637. However, its macro-F1 score decreased from 0.0703 to 0.0658. This indicates that the raw pose features improved balanced accuracy but did not improve class-balanced F1 performance.

The raw-pose model also achieved the lowest total loss, decreasing from 0.02649 for the baseline to 0.02305. It further reduced the L1 loss from 145.6158 to 111.3934 and the TD loss from 0.000763 to 0.000722. By contrast, the action loss of the raw-pose model was slightly higher than that of the baseline (2.2372 vs. 2.2337). Thus, the gain in balanced accuracy did not translate into a lower cross-entropy loss, but it did coincide with lower regularization and total losses.

The pose ST-GCN model showed a different trend. Its balanced accuracy was 0.0849, which was close to the baseline, and its macro-F1 score decreased from 0.0703 to 0.0686. Its action loss was also the highest among the three

models. Therefore, the learned pose embedding did not improve action-prediction performance in the current setting.

Nevertheless, this pose ST-GCN model achieved the lowest TD loss, decreasing from 0.000763 in the baseline to 0.000635. This suggests that the latent pose representation may have helped reduce the TD error. However, this improvement was not reflected in the total loss because the model produced a much larger L1 loss. As a result, the pose ST-GCN model had the highest total loss among the three models.

3.2 Prediction-level Analysis of Baseline and Raw-Pose Predictions

To better understand how the raw-pose model changed the predictions in Table 1, we inspected the full baseline-versus-raw comparison on the held-out match. This comparison contained 149,020 player-frame samples. The raw-pose model achieved a balanced accuracy of 0.1637, compared with 0.0836 for the baseline, and produced 18,114 frames in which raw pose was correct and the baseline was wrong, versus 16,432 frames in which the opposite occurred. Thus, the qualitative evaluation suggests a small net advantage for raw pose, although the gain appears modest and situation-dependent.

Figure 2(a) shows a representative improvement case. For Sergio Gómez, the ground-truth action is *cross*. The baseline predicts *through_pass*, while the raw-pose model predicts the correct action. Because *cross* and *through_pass* are closely related ball-progression actions, this example suggests that explicit posture cues can help disambiguate technically similar actions that are difficult to separate from position and velocity alone.

Figure 2(b) shows a representative failure case. For Phil Foden, the ground-truth action is *down*. Here, the baseline predicts the correct action, whereas the raw-pose model predicts *idle*. This indicates that raw pose does not uniformly improve action recognition and can still suppress simple movement classes that are already captured reasonably well by the baseline state.

Taken together, these examples refine the interpretation of the quantitative results. Raw pose provides a net positive effect, but the gain is uneven across situations. The large number of frames in which both models remain wrong with different predictions also suggests that pose alone is not sufficient to resolve many ambiguous or weakly supervised action labels in the current setting.

4 Discussion

The quantitative and qualitative results together indicate that raw pose provides useful but uneven complementary information to the baseline tracking state. Quantitatively, raw pose improved balanced accuracy and reduced the total loss. Qualitatively, it outperformed the baseline on 18,114 player-frame samples and underperformed on 16,432, yielding a small but consistent net advantage across 149,020 evaluated samples. This pattern supports the view that

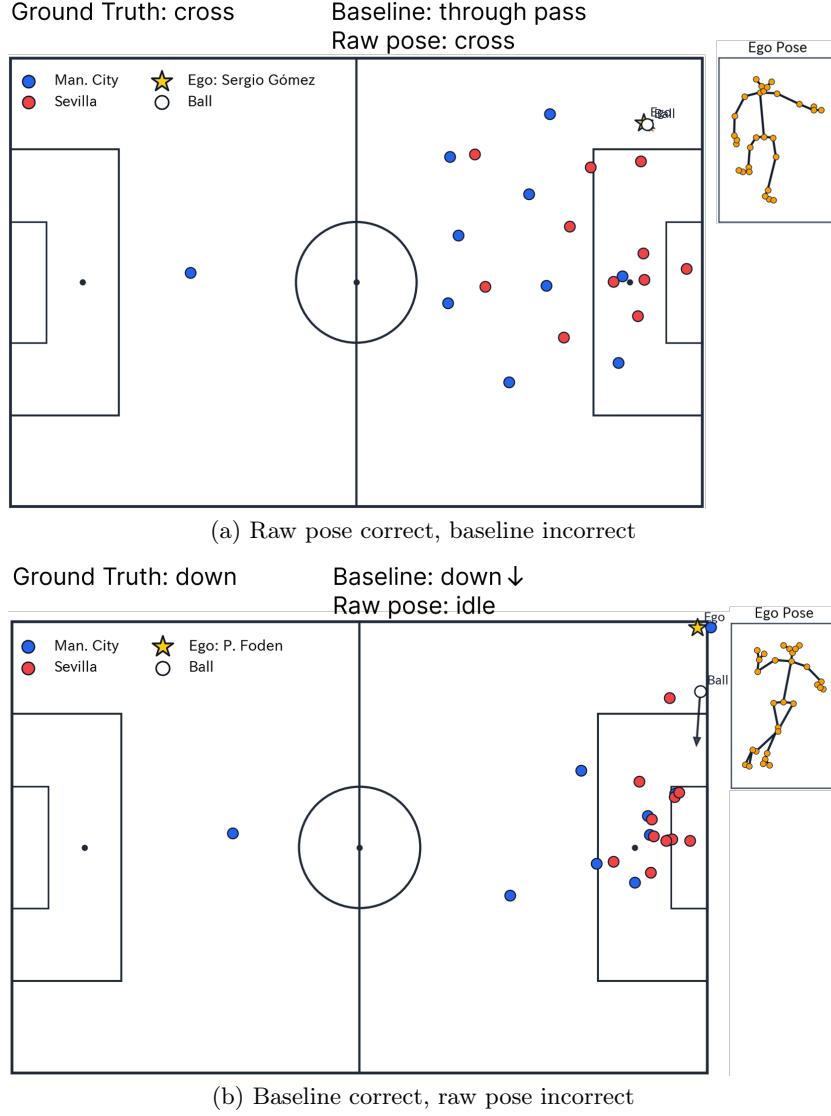


Fig. 2. Representative qualitative comparison between the baseline and raw-pose models on the held-out test match. In (a), the raw-pose model correctly predicts Sergio Gómez’s *cross*, whereas the baseline predicts *through_pass*. In (b), the baseline correctly predicts Phil Foden’s *down* movement, whereas the raw-pose model predicts *idle*. These examples illustrate that raw pose can resolve some action ambiguities, but the gains are not uniform across all player-frame situations.

joint-level posture information can help in action valuation, but that its effect is not large enough to dominate the baseline in every context.

The representative improvement cases suggest where this benefit is likely to come from. In particular, the raw-pose model may help resolve confusions between closely related actions such as *cross* and *through_pass*, when body orientation and preparatory posture provide cues that were not fully encoded by position and velocity alone. Because the raw-pose representation exposed the player-centered 87-dimensional joint vector directly, it may preserve these fine-grained cues more faithfully than a more compressed representation.

However, the raw-pose result should not be interpreted as a uniform improvement across all action-prediction metrics or all situations. The macro-F1 score decreased, the action loss did not decrease, and the baseline still produced many correct predictions that the raw-pose model missed. Moreover, a large number of samples remained difficult for both models. Therefore, the main observed benefit of raw pose was a net improvement in balanced accuracy together with lower total loss, rather than a consistent improvement in class-balanced or probability-based metrics.

The ST-GCN-based latent representation provides an informative contrast. Although it achieved the lowest TD loss, it did not improve the overall action-prediction metrics and produced the highest total loss because of its very large L1 term. A plausible explanation is representational compression: raw pose directly concatenated an averaged 87-dimensional joint vector, whereas ST-GCN compressed a five-frame pose sequence into a 32-dimensional latent code. Under the current data scale, this compression may have discarded action-relevant posture cues before the downstream SARSA model could exploit them reliably.

Another likely factor is supervision. The latent encoder was trained only through the downstream action-valuation objective, without auxiliary pose reconstruction or action-recognition losses. With limited training data and only a short temporal window per 5 Hz bin, this may have made the latent representation unstable and insufficiently discriminative for the action-prediction task. These findings suggest that the current raw-pose features are helpful but still incomplete, while the current latent representation remains under-optimized.

Another limitation is that evaluation was conducted on a single held-out test match. Therefore, the results may depend on match-specific action distributions, tactical contexts, and pose quality. Evaluating the models on a larger number of matches will be necessary to determine whether the observed trends generalize across teams, opponents, and game situations.

5 Conclusions

This paper investigated whether 3D pose information improves MADRL-based football action valuation. In the current setting, the raw-pose model was the most effective of the three representations: it improved balanced accuracy and reduced the total loss relative to the baseline without pose. The qualitative comparison on the held-out match was consistent with this result, suggesting that

explicit joint-level posture information can complement conventional position- and velocity-based tracking features.

At the same time, the benefit of raw pose was not uniform across action classes, and the evaluation was conducted on a single held-out test match. In addition, the current latent ST-GCN representation did not improve the main action-prediction metrics, suggesting that compact pose embeddings remain difficult to optimize effectively under limited supervision and data.

Overall, these findings support the value of pose information for football action valuation while highlighting the importance of representation design. Future work should evaluate the approach on more matches, improve label supervision, and investigate stronger pose-feature fusion and latent-pose learning strategies that can preserve posture cues while yielding more stable gains across different tactical contexts and action categories.

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